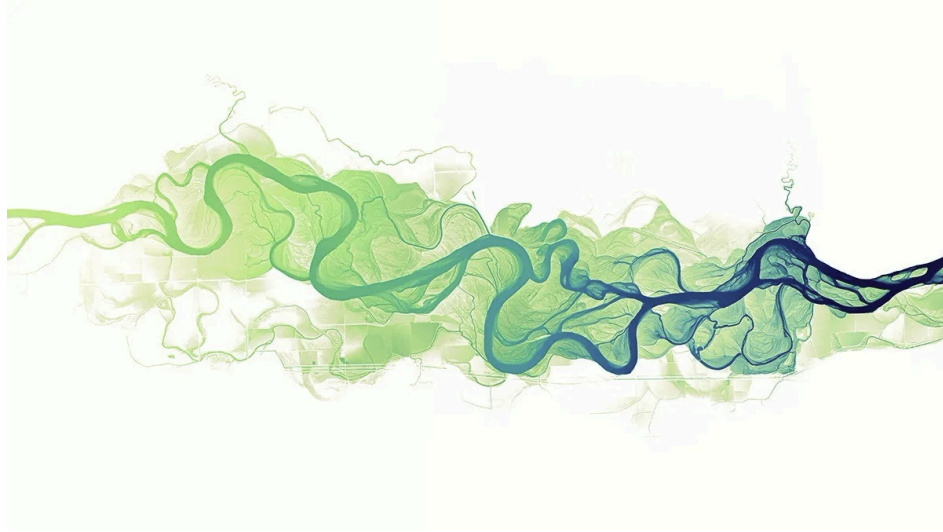


Automated Detection and Morphological Classification of Mast Cells in Bone Marrow Aspirates Using Deep Learning



Bachelor Thesis

by
Florian Vollmer

Bachelors 2023

Submission date: 2026-07-02

Study Direction: Applied Digital Life Science

Supervisor:

Dr. Stefan Glüge

ZHAW Life Sciences und Facility Management, Wädenswil

Cross Corrector:

Prof. Dr. Robert Vorburger

ZHAW Life Sciences und Facility Management, Wädenswil

Imprint

Recommended Citation:

Florian Vollmer (2026). *Automated Detection and Morphological Classification of Mast Cells in Bone Marrow Aspirates Using Deep Learning*

.

Keywords: Deep Learning, Cell Detection, Evaluation Metrics, Digital Lab and Production, Clinical Image Processing

Abstract

English Version

German Version

Acknowledgement

I would like to express my sincere gratitude to Dr. Stefan Glüge for setting up the project for this thesis and for his valuable support and guidance throughout the writing and compilation process of this thesis. My appreciation also goes to Robin Kosche for showing me around the USZ and introducing me to the subject.

Table of Contents

List of Abbreviations	6
1. Introduction	7
1.1. Context and Motivation	7
1.2. Research Questions	7
1.3. Overall Concept	7
1.4. State of the Art	8
2. Methodology	12
2.1. Biology of Mast Cells and Systemic Mastocytosis	12
2.2. Technical	13
3. Results	19
3.1. Model Results and Statistical Evaluation	19
3.2. Quantitative and Qualitative Evaluation	20
3.3. Evaluation Visualization	21
3.4. Real World Usage	21
4. Discussion	22
4.1. Interpretation of Results	22
4.2. Clinical Relevance	22
4.3. Error Analysis	23
5. Conclusion	24
5.1. Summary of Key Findings	24
5.2. Limitations and Challenges	24
5.3. Future Research Directions	24

6. Declaration on AI Assistance	26
7. References	27
8. Lists	29
List of Figures	29
List of Tables	29
9. Attachments	30

List of Abbreviations

- **AI:** Artificial Intelligence
- **AP:** Average Precision
- **CNN:** Convolutional Neural Network
- **COCO:** Common Objects in Context
- **CV:** Cross Validation
- **DoE:** Design of Experiments
- **FN:** False Negative
- **FP:** False Positive
- **GUI:** Graphical User Interface
- **HPC:** High Performance Computing
- **IoU:** Intersection over Union
- **MC:** Mast Cells
- **mAP:** Mean Average Precision
- **RF-DETR:** Roboflow Detection Transformer
- **SM:** Systemic Mastocytosis
- **TN:** True Negative
- **TP:** True Positive
- **USZ:** University Hospital Zurich (Universitätsspital Zürich)
- **WHO:** World Health Organization
- **YOLO:** You Only Look Once

1. Introduction

1.1. Context and Motivation

Systemic mastocytosis (SM) is a rare clonal hematopoietic neoplasm defined by the pathological expansion and accumulation of neoplastic mast cells (MC) in extracutaneous organs, mostly seen in the bone marrow, but also the spleen, liver, and gastrointestinal tract. The major diagnostic criterion for SM is the presence of multifocal MC clusters in the bone marrow and/or extracutaneous organs, while minor criteria include an elevated serum tryptase level, MC expression of CD2, CD25, or CD30, and the presence of activating KIT mutations [1]. Despite these structured criteria, SM is frequently underdiagnosed in routine clinical practice. One of the central reasons for this is that MC, although morphologically distinct, occur only in very small numbers within a given bone marrow aspirate sample, making their manual identification and quantification an exhausting, subjective, and time-consuming task for hematopathologists.

The morphological assessment of MC carries direct diagnostic weight. The abnormal morphology of MC, defined as atypical spindle-shaped cells with hypogranulated cytoplasm and an oval nucleus, represents the first minor SM criterion. At least 25% of all MC must exhibit these morphological features in bone marrow smears or sections for this criterion to be fulfilled [1]. This quantitative threshold requires a reliable and reproducible count of both typical and atypical MC within a sample, a task that is very error-prone when performed manually and at scale.

In clinical practice, the integration of clinical signs and symptoms together with bone marrow morphological, immunophenotypic, and molecular results is required to diagnose SM variants [2]. However, the morphological step remains the entry point into this diagnostic chain. Standardizing and automating this step through computational means therefore has direct implications for diagnostic accuracy and clinical throughput. The University Hospital Zurich (USZ), which operates as the clinical partner for this thesis, has a concrete need for such a tool that could be embedded in its existing cell classification infrastructure.

1.2. Research Questions

The goal of this thesis is to develop and evaluate a deep learning-based pipeline for the automated detection and morphological classification of mast MC in bone marrow aspirate images. From this objective, the following research questions are derived:

1. Which specific performance metrics (e.g., F1-score, Mean Average Precision, or false negative rate) are most critical to ensure the tool's reliability for clinical screening?
2. Is the developed pipeline capable of identifying mast cells in samples with extremely low cell density ("rare event detection") without generating an unmanageable number of false positives?
3. Can an AI-based pipeline reliably distinguish between typical and atypical mast cell phenotypes to support the >25% diagnostic criterion defined by the WHO 2022 classification?

1.3. Overall Concept

The overall system was designed as an end-to-end pipeline including data acquisition, annotation, preprocessing, hyperparameter tuning, model training, model validation, and clinical deployment. Digital bone marrow aspirate images were collected and annotated in close collaboration with

clinical experts at the USZ to produce a labeled training dataset. Initial hyperparameter tuning was applied to ensure optimal starting conditions for training. A stratified k-fold cross-validation strategy was utilized to deliver a robust model evaluation given the limited availability of annotated rare-cell data. A YOLO-family object detection model was selected and fine-tuned for this task. The final pipeline is intended to be accessible either via a user-friendly drag-and-drop interface that automatically presents detection results and a statistical summary of the analyzed sample, meeting the operational requirements set by the USZ or, where feasible, integrated into Cinderella, the cell classification software already in use at the USZ.

1.4. State of the Art

Deep Learning in Digital Hematology

The intersection of deep learning and hematological image analysis has produced noteworthy progress over the past years. Traditionally, the detection and classification of blood cells involved manual microscopic examination by pathologists, a process that is labor-intensive, subjective, and error-prone. The increasing range of diagnostic tests has driven demand for automated, high-throughput solutions that are cost-effective, require a rapid turnaround time, and provide reliable results. Convolutional neural networks (CNNs) and, more recently, transformer-based architectures have addressed this demand by learning to identify and categorize morphological features directly from raw pixel data [3].

Among the detection frameworks applied in this domain, the YOLO (You Only Look Once) family of models has proven particularly effective. Studies applying YOLOv10 and YOLOv11 to the detection and classification of red blood cells, white blood cells, and platelets in microscopic images have demonstrated mean Average Precision scores of up to 93.8%, underscoring the potential of these architectures in supporting hematological analysis and clinical diagnosis [4]. A feature of YOLO-based pipelines is their ability to perform detection and classification in a single forward pass, hence their name “... Look Once”, making them computationally suitable for integration into clinical workflows where inference speed matters. For further information about the mechanisms for YOLO-models see [PA2-FV].

A meta-analysis of over 400 peer-reviewed studies on deep learning in peripheral blood smear analysis confirmed significant advances in automated diagnostic tools, particularly for disease identification in blood cells, while also identifying critical gaps including the limited exploration of differential white blood cell counting, the underdevelopment of automated morphological analysis, and the absence of standardized evaluation criteria [5].

These gaps are directly relevant to the present thesis, which addresses an underexplored subtype of rare hematological cell analysis.

Comparable Work in Cell Detection

A directly relevant contribution in this domain was published by Glüge et al. [6] in collaboration with the University Hospital Zurich. The study evaluated four state-of-the-art CNN architectures on a dataset of 171'374 microscopic single-cell images from bone marrow smears of 945 patients, finding that the best pre-trained model achieved a 53.5% improvement in precision and 7.3% improvement in recall compared to training from scratch, and that out-of-domain pre-training yielded features that transferred well to cell classification tasks.

A review by Ghete et al. [7] covering AI-based cell classification in bone marrow aspirate smears from 2019 to 2024 provides a systematic overview of the broader field. The review found that

performance metrics across published studies are **not directly comparable**, as the majority implemented their own methods on their own datasets, and few were validated on external datasets. This observation directly informed the design choices of the present thesis, which applies stratified cross-validation to ensure statistically robust performance estimates rather than relying on single-split evaluations.

For multi-class white blood cell classification at the subtype level, Diaz et al. [8] applied a YOLOv11-large model with spatial attention to the simultaneous localization and classification of 13 white blood cell subtypes. The model achieved a mAP₅₀₋₉₅ of 77.9% with an overall F1-score of 0.913, with class-wise average precision ranging between 89.3% and 98.2%, confirming reliable recognition even for infrequent cell subtypes. This is particularly instructive for the present work, as the challenge of infrequent subtypes, low absolute count, high inter-class visual similarity, maps directly onto the atypical versus normal MC classification problem.

Finally, Lv et al. [9] reported on the Morphogo system, a CNN-based platform trained on over 2.8 million bone marrow cell images. Evaluated across 508 cases, the system demonstrated the ability to identify over 25 different cell types, achieving a sensitivity of 80.95%, a specificity of 99.48%, and an overall accuracy of 99.01%. The dataset size required to achieve these results underscores the data scarcity challenge that constrains MC-specific models, a bottleneck that the present thesis addresses through targeted augmentation and cross-validation strategies.

Mast Cell Detection and the Clinical Gap

In contrast to blood cell detection, which benefit from large, publicly available datasets, automated MC analysis in bone marrow aspirates remains largely unexplored in the literature. Figure 1 clearly displays this. Deep learning methods including convolutional neural networks, encoder-decoder architectures, and recurrent frameworks have been proposed for the detection and study of MC in the context of mastocytosis, and the spread and development of MC can in principle be studied through microscopic image analysis using AI frameworks [10]. However, validated pipelines specifically targeting morphological classification of normal versus atypical MC in bone marrow aspirates, with direct alignment to WHO diagnostic thresholds, do not yet exist as established clinical tools.

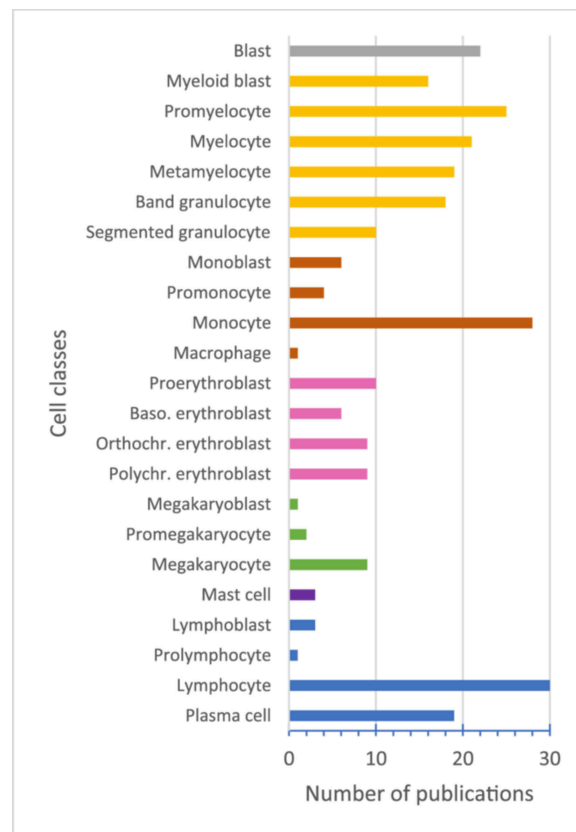


Figure 1: Publication distribution across cell lineages [7]

This gap is clinically significant. The minor diagnostic criteria for SM include the condition that at least a quarter of the MC in the bone marrow exhibit unusual characteristics, such as spindle-shaped (Figure 2) or otherwise abnormal morphology [1]. Meeting or failing to meet this threshold has direct consequences for patient classification and treatment decisions. Manual cell counting is variable between observers and across institutions, and given the low absolute number of MC per sample, even small counting errors can shift a patient across a diagnostic boundary. An automated, reproducible, and transparent counting system therefore addresses a concrete unmet clinical need.

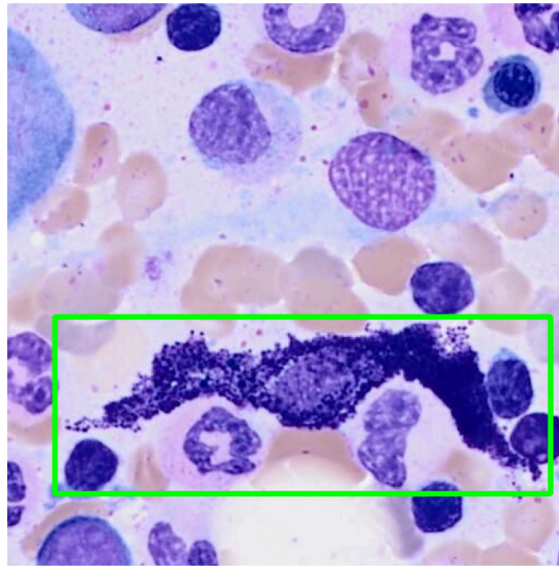


Figure 2: Sample of spindle shaped cell from dataset

Taken together, the existing literature confirms that YOLO-based architectures are well-suited for cell-level detection in hematological microscopy, with performance in the range of 77-94% mAP achievable on well-represented cell types. However, no published study has specifically addressed the binary morphological classification of MC, atypical versus normal, in alignment with WHO diagnostic thresholds for SM. This thesis targets precisely this gap. It builds upon lessons learned from prior object detection experiments in related image analysis tasks [PA2-FV], and applies cross-validation methodology to ensure that reported performance metrics are generalizable rather than optimistically overfitted to a single data split. The clinical ambition of the project is to provide a deployable tool that can be integrated into an existing hospital software environment which distinguishes it from purely academic explorations and places model reliability, interface usability, and system integration at the center of the evaluation criteria.

2. Methodology

2.1. Biology of Mast Cells and Systemic Mastocytosis

Mast Cells: Origin and Function

Mast cells are tissue-resident immune cells that play a central role in the body's innate defense system. They arise in the bone marrow, where maturation is controlled by stem cell factor binding to the receptor c-KIT, as well as by cytokines such as interleukin -3, -4, -9, and -10, which promote both differentiation and proliferation. Once mature, they migrate into neighbouring tissues and take up residence particularly in areas exposed to the external environment such as the gastrointestinal tract, respiratory mucosa, skin, and connective tissue surrounding blood and lymphatic vessels [11].

Structurally, MC contain a nucleus surrounded by hundreds of granules, which serve as storage compartments for chemical mediators. These include histamine, which opens blood vessels and allows fluid and immune cells into tissues; proteases such as tryptase, chymase, and carboxypeptidase, which break down connective tissue to allow for better passage of immune cells; and cytokines and chemokines, which signal immune cells to proliferate, adhere to vessel walls, and move to sites of injury or infection [12].

Beyond their well-known role in allergic reactions, MC also contribute to tissue repair and angiogenesis. MC's synthesize, store, and release hundreds of mediators in response to allergic, environmental, neuroimmune, stress, and toxic triggers, with some of these mediators released independently of degranulation, including angiogenic, fibrotic, and neurotoxic molecules [13]. This functional spectrum makes them relevant not only in allergy and host defense, but across a wide spectrum of inflammatory and neoplastic conditions.

Development of Systemic Mastocytosis

Systemic mastocytosis arises when MC undergo uncontrolled clonal proliferation driven by a somatic gain-of-function mutation. Around 95% of SM cases originate from a point mutation at codon 816 of the KIT gene, where aspartic acid is replaced by valine, referred to as KIT D816V. This substitution results in ligand-independent, constitutive activation of the KIT tyrosine kinase receptor, leading to unregulated MC proliferation, amplified growth, and increased survival [14].

Disease Subtypes and Progression

SM is not a single, uniform entity but a spectrum of disease subtypes that differ in clinical behavior and prognosis. The International Consensus Classification defines the following subtypes in increasing order of severity: indolent SM (incorporating bone marrow mastocytosis), smoldering SM, aggressive SM, SM with an associated hematological neoplasm (SM-AHN), and mast cell leukemia (MCL) where as in the WHO 5th Edition, Bone Marrow Mastocytosis is placed on the top and the ranking shifts by +1 [14].

The majority of adult patients present with indolent SM, which is characterized by mediator-related symptoms, frequent skin involvement, and a near-normal life expectancy. The neoplastic MC burden in the bone marrow remains relatively low, and organ dysfunction is absent. Smoldering SM occupies an intermediate position, with a higher MC burden and a greater risk of progression, but without the organ damage that defines advanced disease.

SM as a Precursor to Other Hematological Neoplasms

One of the most clinically consequential features of SM is its capacity to co-occur with or precede other hematological malignancies. Secondary genetic lesions on top of the KIT mutation, or in cooperation with a particular genetic background, are potentially required for the malignant transformation of indolent SM into more severe disease forms [15]. In SM-AHN, the associated hematological neoplasm, which may include myelodysplastic syndrome, myeloproliferative neoplasm, or acute myeloid leukemia, frequently shares the KIT D816V mutation with the MC compartment, indicating a common clonal origin in a multipotent hematopoietic precursor.

The occurrence of the KIT mutation in bone marrow cell compartments other than MC is associated with progression to more aggressive forms of the disease and poor outcome in indolent SM [16]. This involvement therefore serves as an important prognostic indicator, and its detection at the immunophenotypic level during bone marrow assessment has direct implications for patient management.

2.2. Technical

The technical pipeline of this thesis combines clinical data acquisition, deep learning model selection, iterative training strategy, and a downstream desktop application for clinical use. The work was organized in two clinical data phases. Phase 1 (P1) served as a proof of concept on the first patient cohort and established the baseline training procedure. Phase 2 (P2) introduced the second patient cohort and a more rigorous evaluation strategy including stratified cross-validation, transfer learning from a domain-pretrained checkpoint, and a structured Design of Experiments grid over the dataset composition factors. An overview of the end-to-end pipeline is shown in [?@fig-pipeline-overview](#).

General Data Information

The dataset for this study consists of bone marrow aspirate smears provided by the USZ. Although these samples were originally curated for the manual morphological differentiation between typical and atypical MC, they have been made available for this thesis to serve as the foundation for training and validating the deep learning models. All material originates from patients with confirmed or suspected SM. As a consequence, atypical MC are over-represented relative to a normal distribution within bone marrow aspirate, with direct implications for class balance that are addressed in Section 4.3..

The data is structured into four phases corresponding to four patient sources. Phase 1 (Pos_neg 12241515) and Phase 2 (1224151atypisch_normal) form the primary annotated training material used in this work. Phase 3 (12309915_task7_atypisch_normal) and Phase 4 (12397199_Pos_normal_task8) were reserved for external validation and live inference experiments. All images are stored as JPEG files at the original microscope resolution and are accompanied by YOLO-format text labels containing class index and normalized bounding-box coordinates.

Data Acquisition & Annotation

Bone marrow aspirate smears were prepared and imaged at the USZ using their standard digital microscopy protocol. Annotation was performed in close collaboration with hematopathology

personnel: each candidate cell was manually labeled as either Atypisch (atypical, spindle-shaped or otherwise morphologically abnormal MC) or Normal (typical round, granulated MC). A representative example of a labeled spindle-shaped cell is shown in Figure 2.

In addition to the annotated positive samples, a list of confirmed false-positive negatives was supplied in tabular form (Task_6_1224151_negative.xlsx). These tiles correspond to image regions where earlier model iterations had produced detections that were subsequently confirmed by experts not to contain MC. They were treated as high-quality hard negatives during training (seexxxxxx).

The annotation conventions follow the WHO 5th Edition criteria for atypical MC morphology [1], namely spindle shape, hypogranulation, and oval nuclei. Examples of both classes as encountered in the dataset are provided in ?@fig-class-examples.

Data Preprocessing

Each annotated image was paired with its corresponding YOLO-format label file at load time. A scripted verification step ensured that every image had a matching label and that all class indices fell within the defined two-class range. Images without labels were either rejected (Phase 1) or, for the Phase 2 grid search, optionally added as confirmed negative background tiles to provide the model with examples of cell-free tissue.

All images were uniformly resized to 512×512 pixels at training time, which corresponds to the working resolution of the YOLO11 detection head used in this work. The 512 px choice balances the spatial resolution required to discriminate fine morphological features (granule structure, nuclear shape) against the computational cost of training. Two preprocessing variants were evaluated:

1. **Pre-applied augmentation** — every image was processed once through an augmentation pipeline (horizontal/vertical flip, brightness shift, affine rotation $\pm 15^\circ$) and the result written to disk. Each training epoch then re-used the identical augmented copy.
2. **No-augmentation baseline** — images were copied without modification, providing a clean control for ablation comparison.

Pre-applied augmentation was found to be ineffective in this setting: because each image is augmented exactly once at preprocessing time, every epoch of training sees the same fixed view of the dataset. The effective regularization is therefore exhausted after the first epoch. This observation motivated the recommendation to switch to per-epoch on-the-fly augmentation in future work, where the random transforms are re-sampled at each forward pass through the model. An illustration of this distinction is given in ?@fig-augmentation-strategy.

Model Selection & Architecture

The model family selected for this thesis is YOLO11 (You Only Look Once, version 11) by Ultralytics. YOLO models perform single-stage object detection: a single forward pass through a convolutional backbone, neck, and detection head produces both class probabilities and bounding-box coordinates without a separate region-proposal stage. This design yields the inference speed

required for clinical throughput and allows the model to be deployed on commodity hardware. A schematic of the YOLO11 architecture is shown in [?@fig-yolo-architecture](#).

Two model variants were considered: `yolo11n` (the nano variant, ~2.6 M parameters) and `yolo11l` (the large variant, ~25 M parameters). Initial Phase 1 experiments with the large variant produced poor convergence, attributable to the limited number of annotated samples relative to the model capacity. The nano variant was therefore retained for all subsequent experiments. In Phase 2, the publicly pretrained `yolo11n.pt` weights were replaced as the fine-tuning base with `DL_Model1_FV.pt`, a domain-specific checkpoint provided by the supervisor and pretrained on Phase 1 cell data with one foreground class.

The transition from a one-class pretrained checkpoint to two-class fine-tuning has concrete consequences for which weights transfer and which are reinitialized. Ultralytics constructs a fresh detection head with the new number of classes ($nc=2$) and loads checkpoint weights only where layer names and tensor shapes match. The classification branch of the detection head (`cv3`) outputs nc channels, so its final convolution shape changes from 1 to 2 and its weights are discarded and reinitialized. The bounding-box regression branch (`cv2`) is class-count-independent and transfers cleanly. This is summarized in [?@fig-transfer-mechanics](#).

Model Training & Finetuning Strategy

A 5-fold stratified cross-validation strategy was adopted to compensate for the limited dataset size and to produce performance estimates that reflect the variance inherent to the train/test split. Stratification was applied at the image level using the dominant class label in each image, so that the relative class frequency was preserved across folds. Within each fold, the data was further partitioned into training (approximately 70%), validation (approximately 15%), and held-out test (approximately 15%) splits. The stratified split is illustrated in [?@fig-cv-split](#).

To accelerate training, folds were executed in parallel on the ZHAW HPC cluster (earth-4 partition, two NVIDIA L40S GPUs), with fold assignment to GPU determined by the parity of the fold index. SLURM was used as the workload scheduler. Care was taken to ensure that concurrent jobs did not interfere with each other: temporary fold workspaces and project output directories were namespaced using `$SLURM_JOB_ID` and `$SLURM_JOB_NAME`, respectively. Earlier runs that had collided in shared output paths were identified by their suspiciously identical fold-level results and excluded from the analysis.

Fine-tuning hyperparameters were selected based on an analysis of training behavior across early Phase 2 experiments. The default YOLO learning rate ($lr=0.01$) is calibrated for training from scratch on the COCO dataset and was found to be too aggressive when fine-tuning from `DL_Model1_FV.pt`: the gradients propagating from the reinitialized classification branch during early epochs were large enough to disrupt the pretrained neck features. A $10\times$ reduction to $lr=0.001$ together with freezing the first ten layers of the backbone (`freeze=10`) eliminated the early-epoch collapse observed at the higher learning rate. The final hyperparameter set is given in Table 1.

Table 1: Final fine-tuning hyperparameters used for all Phase 2 grid runs.

Hyperparameter	Value	Notes
<code>lr0</code>	0.001	Reduced 10× from YOLO default
<code>lrf</code>	0.01	Final LR = <code>lr0</code> × <code>lrf</code> = 1e-5 (cosine schedule)
<code>cos_lr</code>	True	Cosine annealing
<code>freeze</code>	10	First 10 layers (backbone) frozen
<code>epochs</code>	5000	Upper bound; early stopping active
<code>patience</code>	100	Epochs without val improvement before stop
<code>batch</code>	32	Per-GPU batch size
<code>imgsz</code>	512	Input image size
<code>mosaic</code>	0.0	Disabled (mosaic scrambles single-cell crops)
<code>cfg</code>	not used	Tuner-derived hyperparameters cause forgetting

The use of YOLO’s built-in tuner (`model.tune()`) was investigated and rejected for fine-tuning. Although the tuner produces hyperparameters that are well-calibrated for training from scratch, applying them to a fine-tuning run with a domain-pretrained checkpoint resulted in catastrophic forgetting: validation classification loss climbed to ~5.8 within the first three to six epochs (typical range 0.5–1.5), and all five folds collapsed to a near-identical degenerate solution.

In addition to the model hyperparameters, two dataset-composition factors were varied across runs:

- **FP_NEG_OVERSAMPLE** — the number of times each FP-negative tile is repeated in the training file list (values tested: 1, 2, 3).
- **BG_RATIO** — the number of unlabeled background tiles added per annotated image as confirmed negatives (values tested: 0, 1, 2, 3).

These two factors were swept in a full 3×4 factorial Design of Experiments to identify whether either had a meaningful effect on test-set performance.

Model Testing

Each fold was evaluated on its held-out test split using the best checkpoint produced during training (`best.pt`, selected by validation mAP50). Test-set evaluation was performed with `model.val(split='test')`, which computes precision, recall, mAP50, and mAP50-95 at the standard COCO IoU thresholds. Validation metrics observed during training were not used for model comparison; only the held-out test split is reported.

For the deployment pipeline, the operating confidence threshold was tuned on the validation curves rather than fixed at the YOLO default. The thesis-evaluated thresholds for the deployed model are `conf=0.58` and `iou=0.45`. A separate research configuration uses `conf=0.276` and `conf=0.25` for sensitivity analyses where the full confidence distribution should be retained. The downstream decision of where to set the operating threshold is informed by the per-image

confidence histogram, which is collected during mass inference on unseen images and used to identify the cutoff above which detections are clinically reliable.

Evaluation Metrics

Detection performance was assessed with the metrics standard to the YOLO/COCO evaluation tradition:

- **Precision** — fraction of model detections that match a ground-truth annotation ($TP/(TP+FP)$).
- **Recall** — fraction of ground-truth annotations that are correctly detected ($TP/(TP+FN)$).
- **mAP50** — mean average precision averaged over both classes at an IoU threshold of 0.50. Sensitive to whether a detection exists in approximately the right location.
- **mAP50-95** — mean average precision averaged over IoU thresholds from 0.50 to 0.95 in steps of 0.05. Sensitive to localisation quality in addition to detection presence.

For this work, recall is the metric of greatest clinical interest. Missing an atypical MC has a more severe clinical consequence than producing a false-positive flag, because the former can result in an undiagnosed case of SM whereas the latter is corrected by expert review of the flagged region. The reported test-set means and standard deviations are computed across the five CV folds and serve as the primary basis for comparing experimental configurations.

UI-Creation

A desktop application (`MastCellDetector`) was developed to make the trained model usable in a clinical setting without dependence on a server or notebook environment. The application is implemented in Python with PySide6 (Qt 6 bindings) and is packaged as a self-contained binary using PyInstaller. The high-level layout consists of a sidebar with source loading, hardware information, model selection, inference controls, and retraining controls; a central gallery view; and an annotation editor reachable by activating a thumbnail. A screenshot of the gallery view is shown in [?@fig-app-gallery](#).

Several engineering features were implemented to support real-world clinical folders, which can contain thousands of high-resolution images:

- **Chunked inference** — images are processed in user-configurable chunks (default 250 per `predict()` call). This bounds peak GPU and RAM usage independently of folder size, allowing folders of 20'000+ images to be processed on commodity hardware. Each chunk persists its results to a per-folder `.dapp_meta.json` metadata file before the next chunk begins; an interrupted run can therefore be resumed without re-processing already-labeled images.
- **Annotation editor** — clinicians can review, correct, add, or delete bounding boxes on any image. Edited images are flagged in the metadata file and are never overwritten by subsequent automated inference. The editor view is shown in [?@fig-app-editor](#).
- **In-application fine-tuning** — the user can trigger a fine-tuning run on the locally edited images using the currently loaded model as the base. Training batch size is automatically set to one third of the inference batch size to account for the higher VRAM requirement of backpropagation, gradients, and optimizer states.

- **Persistent settings** — confidence threshold, batch size, chunk size, retraining epoch count, and last-used model path are stored across sessions using the Qt QSettings mechanism.
- **Hardware abstraction** — the application detects available CUDA, ROCm, MPS (Apple Silicon), or CPU backends at startup and recommends an appropriate batch size based on the detected VRAM.

Implementation in USZ Infrastructure

Two integration paths into the USZ infrastructure were considered: (i) deployment of the standalone MastCellDetector desktop application as a clinician-operated review tool, and (ii) integration of the trained weights into Cinderella, the cell classification software already in use at the USZ. The standalone application was the primary deliverable of this thesis as it could be developed and validated independently of the Cinderella codebase. Its drag-and-drop folder interface, low hardware requirements, and offline operation make it suitable for use on existing hospital workstations without modification of the broader IT infrastructure.

Direct Cinderella integration is treated as future work. The trained YOLO11 weights are stored in a portable .pt format and the inference code path is isolated in a single module, which should facilitate eventual integration without requiring the Cinderella codebase to depend on PySide6 or the desktop UI components.

3. Results

3.1. Model Results and Statistical Evaluation

The Phase 2 results are organized into two groups: (i) the iterative single-configuration runs (P2-A through P2-H) that established the fine-tuning recipe, and (ii) the full 3×4 Design of Experiments grid that systematically varied `FP_NEG_OVERSAMPLE` and `BG_RATIO` while holding the remaining hyperparameters fixed at the values established by P2-H. All metrics reported in this section are test-set (held-out split) values, computed with `model.val(split='test')` on the best checkpoint of each fold.

Table 2 summarizes the mean test-set performance across the principal Phase 2 runs.

Table 2: Test-set summary of the principal Phase 2 single-configuration runs. P2-H represents the configuration adopted as the reference for the DoE grid.

Run	Base Model	FP×	BGr	lr0	freeze	mAP50 (mean ± std)	Recall
P2-A	yolo11n	×1	0	0.01	—	0.580 ± 0.172	0.617
P2-B	yolo11n (mod.)	×1	0	0.01	—	0.614 ± 0.155	0.667
P2-C	DL_Modell_FV	×1	0	0.01	—	0.469 ± 0.035	0.397
P2-D	DL_Modell_FV (mod.)	×1	0	0.01	—	0.590 ± 0.090	0.655
P2-G	DL_Modell_FV	×1	2	0.01	—	0.530 ± 0.090	0.484
P2-H	DL_Modell_FV	×1	0	0.001	10	0.604 ± 0.194	0.537

Table 3 presents the complete DoE grid of `FP_NEG_OVERSAMPLE` × `BG_RATIO`. The single corner cell at `fp=3`, `bgr=3` (320709) failed to complete and is marked accordingly.

Table 3: Test-set results of the DoE grid. All runs use `lr0=0.001`, `freeze=10`, `DL_Model1_FV.pt`. mAP50 is the mean across the five CV folds.

Config	FP×	BGr	mAP50	mAP50-95	Recall	Precision
fp1_bgr0	×1	0	0.604	0.494	0.537	0.934
fp1_bgr1	×1	1	0.562	0.438	0.531	0.911
fp1_bgr2	×1	2	0.564	0.433	0.523	0.828
fp1_bgr3	×1	3	0.591	0.441	0.518	0.820
fp2_bgr0	×2	0	0.581	0.446	0.638	0.726
fp2_bgr1	×2	1	0.565	0.426	0.523	0.920
fp2_bgr2	×2	2	0.592	0.446	0.637	0.831
fp2_bgr3	×2	3	0.596	0.467	0.539	0.792
fp3_bgr0	×3	0	0.548	0.407	0.537	0.908
fp3_bgr1	×3	1	0.566	0.434	0.546	0.798
fp3_bgr2	×3	2	0.572	0.422	0.526	0.925
fp3_bgr3	×3	3	— failed —	—	—	—

A heat-map representation of mAP50 across the grid is given in `?@fig-doe-heatmap` and visualizes the absence of a strong main effect for either factor.

3.2. Quantitative and Qualitative Evaluation

The single dominant feature of the Phase 2 results is the persistent fold-level heterogeneity. Across all eleven completed DoE configurations, fold 3 produces a markedly higher mAP50 (range 0.88–0.94) and recall (range 0.80–0.95) than folds 1, 2, 4, and 5, which cluster in the range 0.44–0.61 mAP50 and 0.42–0.49 recall. This pattern is illustrated in `?@fig-fold-variance`.

The cross-fold standard deviation (~0.15–0.21 across all configurations) exceeds the differences between configurations (mAP50 spread 0.548–0.604, range 0.056). No configuration is therefore statistically distinguishable from the P2-H baseline once fold variance is accounted for.

The factor-level effects, despite being small in absolute terms, are nonetheless internally consistent:

- **BG_RATIO** has a uniformly negative effect on mAP50. Within every level of `FP_NEG_OVERSAMPLE`, the `bgr=0` cell achieves the best or tied-best mAP50, and each step up in background ratio degrades the metric.
- **FP_NEG_OVERSAMPLE** plateaus above ×1. The ×2 setting yields the highest reported recall (0.638) but inspection of the per-fold values shows this is driven almost entirely by a single fold-2 outlier (recall 0.97). Without that fold, ×2 is indistinguishable from ×1. The ×3 setting underperforms both.

`?@fig-fold-recall` complements the mAP50 visualization with the corresponding recall pattern.

3.3. Evaluation Visualization

?@fig-confusion-matrix shows the normalized confusion matrix of the best-performing configuration (P2-H, fold 3) at the deployment threshold of $\text{conf}=0.58$. ?@fig-pr-curves plots the per-class precision–recall curves used to select this threshold, with the operating point marked.

?@fig-conf-distribution shows the confidence-score distribution observed during mass inference on unseen images and motivates the choice of the operating threshold: detections above approximately 0.58 are dominated by true positives, while the long tail below this value is enriched in false positives.

3.4. Real World Usage

Tests With Unseen Data

A mass inference run was performed on the full set of unseen Phase 1 images, with all images that participated in training filtered out by filename. Each detection was retained together with its confidence score; the resulting `confidence_distribution.csv` file underpins ?@fig-conf-distribution. The mass inference further allowed visual inspection of detections in clinically realistic conditions and was used to verify that the model’s failure modes (most prominently, false positives on cell-like staining artifacts) match the modes anticipated from the training-set behaviour.

The desktop application was used end-to-end on a folder of unseen images comparable in size to a routine clinical batch. The chunked inference, gallery filtering, and annotation-editor workflow performed as designed: clinicians reviewed flagged images, corrected false positives by deleting boxes, and added missed cells through the annotation editor. The corrected images were then used as the seed for a short in-application fine-tuning run, demonstrating the active-learning loop end-to-end. A representative screenshot of a clinically-reviewed folder is shown in ?@fig-app-clinical-use.

4. Discussion

4.1. Interpretation of Results

The principal finding of this work is that, within the explored design space, neither `FP_NEG_OVERSAMPLE` nor `BG_RATIO` produces a meaningful main effect on test-set performance once cross-fold variance is taken into account. The DoE grid therefore failed to identify a configuration that materially outperforms the P2-H reference ($\text{mAP}_{50} = 0.604 \pm 0.194$). Instead, the dominant source of variance is the fold split itself: a single fold (fold 3) consistently produces metrics 0.3–0.5 points above the remaining four folds, and this pattern persists regardless of which dataset-composition factors are varied.

This pattern admits two interpretations. The first is that the test images that happen to fall into fold 3 are systematically easier than those in the other folds — for example, because they contain larger or more clearly-stained cells. The second is that the training images excluded from fold 3 contain examples that are particularly important for generalising to the other test folds, and that their absence in fold 3’s training set does not hurt because the test set itself is easier. In either case, the implication is that the dataset is too small for the train/test split to be statistically representative, and that data-composition factors of the kind explored in the DoE cannot resolve this through volume alone.

The comparison between the YOLO-default and reduced-learning-rate configurations is the one robust effect that the experimental campaign did identify. P2-C, which fine-tuned `DL_Model1_FV.pt` at the YOLO default learning rate ($\text{lr} = 0.01$), produced a mean recall of only 0.397 and was demonstrably overwriting the pretrained features. The $10\times$ reduction to $\text{lr} = 0.001$ together with `freeze=10` recovered both convergence stability (no fold-collapse) and a recall of 0.537 at the same level of precision. This configuration is the recommended fine-tuning recipe for `DL_Model1_FV.pt` going forward.

The failure of the YOLO built-in tuner (P2-E) further demonstrates that hyperparameters optimized for from-scratch training do not generalize to fine-tuning. The tuner search produced a `cfg` file that was internally consistent but, when applied to a domain-pretrained checkpoint, caused all five folds to collapse to the same degenerate solution within a few epochs. This observation is consistent with the transfer-learning analysis of the detection head and suggests that any future tuner-based search should be performed with the pretrained checkpoint and the `freeze` setting in place.

4.2. Clinical Relevance

The clinical motivation for this work is the WHO 5th Edition criterion that at least 25% of the MC in a bone marrow aspirate must exhibit atypical morphology to fulfill the relevant minor diagnostic criterion for SM [1]. Crossing or failing to cross this threshold has direct consequences for patient classification. From this clinical framing, two performance characteristics of the developed pipeline are particularly important.

First, recall on the *Atypisch* class must be high. Missing an atypical MC understates the numerator of the diagnostic ratio and can lead to a false-negative SM screen. The recall-first orientation of the threshold selection (`conf=0.58` was chosen at the inflection point of the precision–recall curve, deliberately on the recall-favorable side) reflects this priority.

Second, recall on the Normal class must also be reliable. The diagnostic threshold is a ratio, not an absolute count, so under-detection of Normal MC inflates the apparent atypical fraction. As discussed in Section 4.3., this is particularly consequential when a patient who does not have SM is screened with a model trained predominantly on SM-positive material.

The desktop application addresses these clinical requirements through a workflow that does not require the model to be perfectly autonomous. Detections are surfaced to the clinician through filterable galleries; uncertain detections (low-confidence tab) and confirmed errors (edited tab) are tracked separately; and corrections feed back into a fine-tuning loop. This brings the pipeline closer to the active-learning paradigm than to a fully autonomous diagnostic engine, which is the appropriate level of automation for a clinical screening tool subject to expert review.

4.3. Error Analysis

The error structure of the pipeline can be decomposed into three categories: dataset-induced errors, fold-split-induced errors, and intrinsic model errors.

Dataset-induced errors. The training material originates exclusively from patients with confirmed or suspected SM. As a consequence, atypical MC are over-represented relative to a reference distribution sampled from a general population. A model trained on this material learns the prior that mast cells in bone marrow are likely atypical. Two consequences follow. First, recall on the Atypisch class is good by construction because the class is well-represented. Second, recall on the Normal class is comparatively weak because the model has rarely seen images dominated by Normal MC. Under-detection of Normal MC then biases the Atypisch/Normal ratio upward and risks producing false SM flags on borderline non-SM patients. This is the most consequential systematic error of the present pipeline and should be addressed by inclusion of bone marrow material from non-SM patients in any future training round.

Fold-split-induced errors. As discussed under xxxxxxxxxxxxxx and the Results, the cross-fold standard deviation dominates the configuration-level differences observed in the DoE. This is a small-dataset symptom rather than a defect of the model or training procedure, and it can only be resolved by either (i) increasing the absolute number of annotated images, or (ii) adopting a leave-one-patient-out cross-validation scheme that explicitly stresses inter-patient generalisation rather than intra-patient interpolation.

Intrinsic model errors. Visual inspection of the false-positive detections in the unseen mass-inference set shows that the most common failure mode is detection on cell-like staining artifacts (clumped granules, debris, or partially-stained nuclei from non-MC lineages). These tiles are exactly the material that has been collected as FP negatives for the next training round, which closes the loop on this error category through the active-learning workflow described above.

5. Conclusion

5.1. Summary of Key Findings

This thesis developed and evaluated an end-to-end pipeline for the automated detection and morphological classification of mast cells in bone marrow aspirate images, from data acquisition and annotation through model training and clinical deployment. The principal findings are:

1. Fine-tuning a YOLO11 model from a domain-pretrained checkpoint (DL_Model1_FV.pt) requires a substantially reduced learning rate (`lr0=0.001`) and a frozen backbone (`freeze=10`) to avoid early-epoch overwriting of the pretrained features. Applying YOLO's built-in hyperparameter tuner to this fine-tuning task is counterproductive.
2. A 3×4 DoE grid over FP_NEG_OVERSAMPLE and BG_RATIO produced no configuration that materially outperformed the reference setting (`fp=1, bgr=0`). The total mAP50 spread across the eleven completed grid cells (0.548–0.604) is smaller than the within-cell cross-fold standard deviation (~0.19). Background tiles have a uniformly negative effect on test mAP50 within the explored range; FP-negative oversampling beyond ×1 produces marginal returns at best.
3. The dominant source of performance variance across the experimental campaign is the train/test fold split. A single fold (fold 3) consistently outperforms the remaining four folds across all configurations, indicating that the dataset is too small for the splits to be individually representative and that data-composition factors cannot compensate for this.
4. A clinically usable desktop application was developed and validated end-to-end on unseen data. The application implements chunked inference for arbitrary folder sizes, an annotation editor with persistent edit tracking, and an in-application fine-tuning loop that closes the active-learning cycle on a single workstation.

5.2. Limitations and Challenges

The most substantial limitation of this work is the size and composition of the training material. All training data originates from SM-positive or SM-suspected patients, which biases the learned class distribution toward the Atypisch class and limits the model's ability to generalize to non-SM patients. This is a clinically meaningful out-of-distribution problem that cannot be resolved through hyperparameter tuning or data augmentation alone.

A related limitation is the absence of an external validation cohort. All metrics reported in this thesis are computed on held-out splits that originate from the same image distribution as the training material. The performance of the pipeline on bone marrow aspirate images from a different clinical site, microscope setup, or staining protocol has not been measured.

A third limitation, technical rather than clinical, is the absence of on-the-fly augmentation in the runs that constitute the formal evaluation. The pre-applied augmentation strategy used in early experiments was found to be ineffective and was disabled for the DoE; per-epoch random augmentation was identified as the appropriate replacement but was not yet evaluated within the timeframe of this thesis.

5.3. Future Research Directions

Three directions for follow-on work are immediately implied by the findings of this thesis:

1. **Per-epoch on-the-fly augmentation.** The single configuration change most likely to close the val/test gap and reduce inter-fold variance is to enable per-epoch random augmentation (mosaic=0.0 together with horizontal/vertical flips and HSV jitter). The DoE has shown that the recall ceiling outside fold 3 is not attributable to dataset composition, which makes augmentation-based regularization the natural next intervention.
2. **Inclusion of non-SM material.** Adding bone marrow aspirate images from patients without SM to the training distribution would directly address the most consequential systematic error of the present pipeline. This is also the precondition for any external validation effort that targets non-SM populations.
3. **Leave-one-patient-out cross-validation.** Replacing the current image-level stratified CV with a patient-level scheme would explicitly stress inter-patient generalisation and would produce performance estimates more directly comparable to the clinical setting in which a new patient's slides are scored against a model trained on prior patients.

Beyond these immediate steps, integration of the trained weights into the existing Cinderella platform at the USZ remains the longer-term deployment objective. The standalone desktop application developed here can serve as the validation harness and as the fallback deployment path if direct Cinderella integration is delayed.

6. Declaration on AI Assistance

The author declares that the empirical research, data analysis, experimental design, and quantitative results presented in this thesis were generated independently and strictly remain the intellectual property of the author.

Artificial intelligence (AI) assistance was utilized solely for the purpose of language refinement, stylistic editing, and optimization of academic reading flow. Specifically, the AI model (Gemini 2.5 Pro & Claude Sonnet 4.5) were employed to translate text, correct grammar, and restructure sentences into the required third-person past tense and formal academic style.

No core scientific content, model selection decisions, hyperparameter choices, or interpretation of results were delegated to or generated by the AI model.

7. References

- [1] P. Valent *et al.*, “Updated diagnostic criteria and classification of mast cell disorders: A consensus proposal,” *HemaSphere*, vol. 5, no. 11, p. e646, Oct. 2021, doi: 10.1097/HS9.0000000000000646.
- [2] C. Ustun, F. Keklik Karadag, M. A. Linden, P. Valent, and C. Akin, “Systemic mastocytosis: Current status and challenges in 2024,” *Blood Advances*, vol. 9, no. 9, pp. 2048–2062, Apr. 2025, doi: 10.1182/bloodadvances.2024012612.
- [3] S. Choudhary, S. Kumar, P. Siddhaarth, *et al.*, “Advancing blood cell detection and classification: Performance evaluation of modern deep learning models,” *BMC Medical Informatics and Decision Making*, vol. 25, no. 207, 2025, doi: 10.1186/s12911-025-03027-2.
- [4] H. Sazak and M. Kotan, “Automated blood cell detection and classification in microscopic images using YOLOv11 and optimized weights,” *Diagnostics*, vol. 15, no. 1, 2025, doi: 10.3390/diagnostics15010022.
- [5] I. N. Margret and K. Rajakumar, “Deep learning techniques for analyzing peripheral blood smears: A meta-analysis,” *Neural Computing and Applications*, vol. 37, pp. 18039–18065, 2025, doi: 10.1007/s00521-025-11401-4.
- [6] S. Glüge, S. Balabanov, V. H. Koelzer, and T. Ott, “Evaluation of deep learning training strategies for the classification of bone marrow cell images,” *Computer Methods and Programs in Biomedicine*, vol. 243, p. 107924, 2024, doi: <https://doi.org/10.1016/j.cmpb.2023.107924>.
- [7] T. Ghete *et al.*, “Models for the marrow: A comprehensive review of AI-based cell classification methods and malignancy detection in bone marrow aspirate smears,” *HemaSphere*, vol. 8, no. 12, p. e70048, 2024, doi: <https://doi.org/10.1002/hem3.70048>.
- [8] J. Diaz, A. Deb, A. Lyubimova, L. Santos-Carrion, C. Romagnoli, and J. Barrientos, “Machine learning-based detection model for leukemia cells in peripheral blood smears using YOLOv11-large,” *Blood*, vol. 146, p. 6123, 2025, doi: <https://doi.org/10.1182/blood-2025-6123>.
- [9] Z. Lv, X. Cao, X. Jin, *et al.*, “High-accuracy morphological identification of bone marrow cells using deep learning-based morpho system,” *Scientific Reports*, vol. 13, no. 13364, 2023, doi: 10.1038/s41598-023-40424-x.
- [10] V. Srilakshmi, K. S. Chakradhar, K. Suneetha, C. S. Bindu, N. Yamsani, and K. R. Madhavi, “Artificial intelligence for detecting prevalence of indolent mastocytosis,” in *Proceedings of the 14th international conference on soft computing and pattern recognition (SoCPaR 2022)*, A. Abraham, T. Hanne, N. Gandhi, P. Manghirmalani Mishra, A. Bajaj, and P. Siarry, Eds., Cham: Springer Nature Switzerland, 2023, pp. 33–43.
- [11] K. Amin, “The role of mast cells in allergic inflammation,” *Respiratory Medicine*, vol. 106, no. 1, pp. 9–14, Jan. 2012, doi: 10.1016/j.rmed.2011.09.007.

-
- [12] Cleveland Clinic, “Mast cells: Anatomy, function & diseases.” Accessed: Apr. 14, 2026. [Online]. Available: <https://my.clevelandclinic.org/health/body/mast-cells>
- [13] T. C. Theoharides *et al.*, “Mast cells and inflammation,” *Biochimica et Biophysica Acta (BBA) - Molecular Basis of Disease*, vol. 1822, no. 1, pp. 21–33, 2012, doi: <https://doi.org/10.1016/j.bbadis.2010.12.014>.
- [14] R. Alyamany *et al.*, “Unraveling the rare entity of KIT D816V-negative systemic mastocytosis,” *Journal of Hematology*, vol. 13, no. 3, 2024, Available: <https://www.thejh.org/index.php/jh/article/view/1279>
- [15] A. C. Garcia-Montero *et al.*, “KIT D816V–mutated bone marrow mesenchymal stem cells in indolent systemic mastocytosis are associated with disease progression,” *Blood*, vol. 127, no. 6, pp. 761–768, Feb. 2016, doi: [10.1182/blood-2015-07-655100](https://doi.org/10.1182/blood-2015-07-655100).
- [16] C. Teodosio *et al.*, “An immature immunophenotype of bone marrow mast cells predicts for multilineage D816V KIT mutation in systemic mastocytosis,” *Leukemia*, vol. 26, no. 5, pp. 951–958, May 2012, doi: [10.1038/leu.2011.293](https://doi.org/10.1038/leu.2011.293).
- [9] Gemini 2.5 Pro. (n.d.). Google Gemini. Retrieved October 1, 2025, from <https://gemini.google.com/app>
- [10] DeepL Translate: The world’s most accurate translator. (n.d.). <https://www.deepl.com/>
- [11] F. Vollmer, „GitHub Repository: Glass Defect Detection-Evaluating-Object-Models“, GitHub, 1. December 2025. <https://github.zhaw.ch/vollmflo/Glass-Defect-Detection-Evaluating-Object-Detection-Models>

8. Lists

List of Figures

Figure 1	Publication distribution across cell lineages [7]	10
Figure 2	Sample of spindel shaped cell from dataset	11
Cover Picture[9]		

List of Tables

Table 1	Final fine-tuning hyperparameters used for all Phase 2 grid runs.	16
Table 2	Test-set summary of the principal Phase 2 single-configuration runs. P2-H represents the configuration adopted as the reference for the DoE grid.	19
Table 3	Test-set results of the DoE grid. All runs use <code>lr0=0.001</code> , <code>freeze=10</code> , <code>DL_Model1_FV.pt</code> . mAP50 is the mean across the five CV folds.	20

9. Attachments